



Demand and Supply

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Individual Demand

Individual demand refers to the quantity of a good or service that a single consumer is willing and able to buy at different prices during a specific period of time. It is not just about wanting a product, but also having the ability to pay for it. Individual demand usually follows the law of demand, which means that when the price of a product decreases, the consumer buys more of it, and when the price increases, the consumer buys less. The individual demand of a consumer depends on many factors such as the price of the product, the income of the consumer, their tastes and preferences, the prices of related goods, and their expectations about the future.

- **Concept of Individual Demand**
- **Desire + Ability to Pay**
 - Demand is not just wanting something. You must also have **money to buy it**. For example, wanting a car but not having money is not demand.
- **Inverse Relationship with Price**
 - Normally, if the price goes **down**, you will buy **more**.
 - If the price goes **up**, you will buy **less**.
 - This is explained by the **Law of Demand**.

- **Depends on Several Factors**
- **Price of the good** → higher price = lower demand.
- **Income of the consumer** → higher income = more demand (usually).
- **Tastes and preferences** → if you like something more, demand increases.
- **Prices of related goods** → demand can change if substitute or complementary goods' prices change.
- **Future expectations** → if you expect prices to rise, you may demand more today.

Law of Demand:

Definition

The **Law of Demand** states that, *other things remaining the same (ceteris paribus), the quantity of a commodity demanded increases when its price falls, and decreases when its price rises.*

Explanation

When the **price of a good falls**, it becomes **cheaper**, so consumers buy **more** of it.

When the **price rises**, the good becomes **expensive**, so consumers buy **less**.

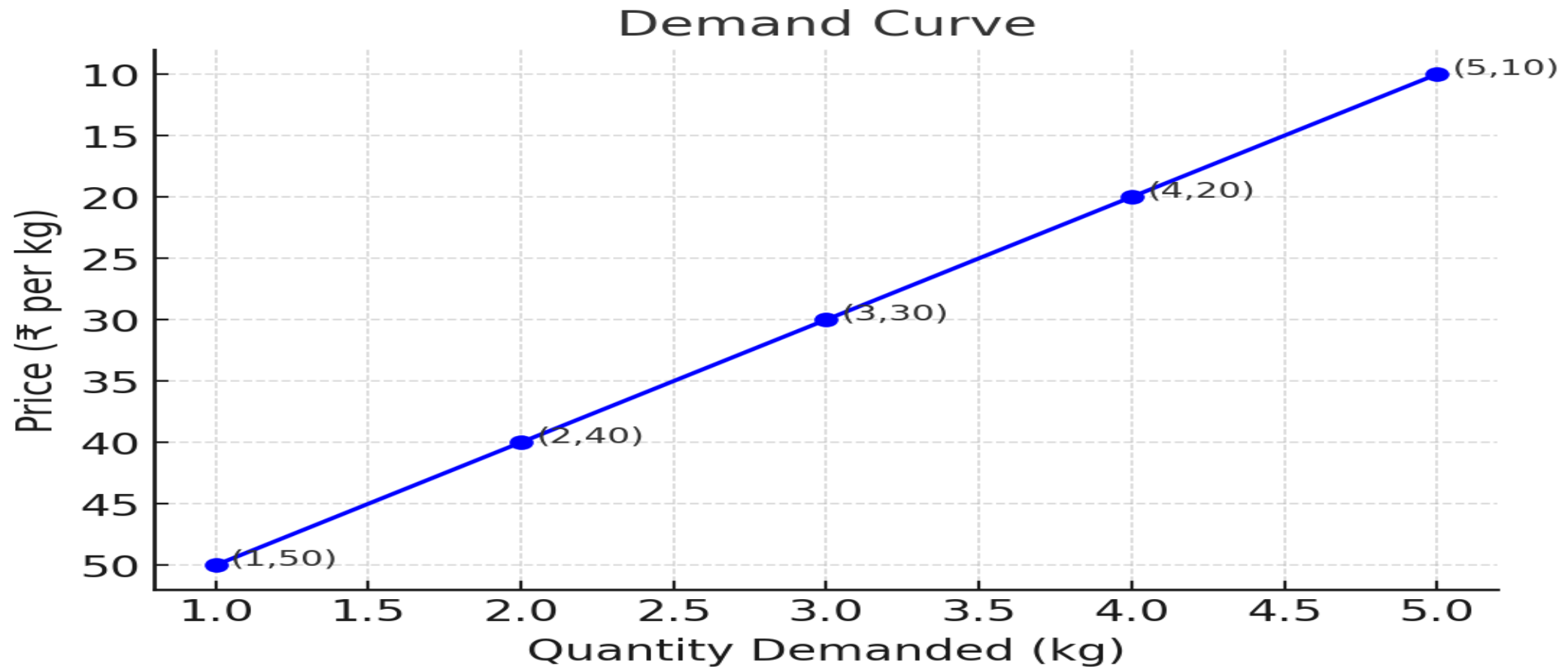
This happens because of two main reasons:

Substitution Effect – When the price of a good falls, people may replace costlier goods with this cheaper one.

Income Effect – A fall in price increases the real income (purchasing power), so people buy more.

Assumptions of the Law of Demand

- The law holds true only when the following conditions remain constant:
- **Income of the consumer remains constant.**
- **Tastes, habits, and preferences do not change.**
- **Prices of related goods (substitutes & complements) remain the same.**
- **Consumers do not expect future price changes.**
- **Population size and distribution of income remain constant.**
- The commodity is a **normal good** (prestige good).



Here's a **clear demand curve** showing the **inverse relationship** between price and quantity demanded: as price falls, demand rises.

Market Demand

Market demand means the total quantity of a commodity that **all consumers in a market** are willing and able to buy at different prices during a given period of time. While **individual demand** shows how much a single person will buy, **market demand** is the sum of the demand of all consumers in the market. Just like individual demand, market demand also generally follows the **law of demand** – when price decreases, total demand increases, and when price increases, total demand decreases.

- **Aggregation of Individual Demands:**

Market demand is obtained by **adding together the individual demands** of all consumers at each price level. This process is called **horizontal summation** of individual demands.

- For example, suppose there are two consumers, A and B. If at a price of ₹30, A demands 3 kg of apples and B demands 4 kg, then the **market demand** is 7 kg. Similarly, for every price level, we add the demands of A and B to get the total demand in the market.

Market demand = **Total demand of all buyers in the market**, and it is calculated by **adding up individual demands at each price**.

Determinants of Demand

1. Price of the Commodity

- When the price of a product falls, people usually buy more of it.
- When the price rises, people buy less.
- (Inverse relationship – explained by the Law of Demand).

2. Income of Consumers

- If people's income increases, they can buy more goods (demand rises).
- If income falls, demand decreases.
- Example: With a salary hike, you may eat out more often.

3. Prices of Related Goods

Substitutes: If the price of tea rises, demand for coffee may increase (because people switch).

Complements: If car prices fall, demand for petrol may rise (because more cars are bought).

4. Tastes and Preferences

- If a product becomes fashionable or healthier, demand increases.
- If people lose interest (e.g., outdated tech), demand decreases.

5. Expectations about Future

- If people expect prices to rise soon, they may buy more today.
- If they expect prices to fall, they may postpone purchases.

6. Population and Demographics

- Larger population = higher demand.
- Age, gender, and lifestyle also matter (e.g., more young people = more demand for gadgets).

Law of Supply

The **Law of Supply** states that, **other things being equal**, the quantity of a commodity supplied increases when its price increases, and decreases when its price falls.

This happens because higher prices make production more **profitable**, so producers are willing to sell more.

- Conversely, when prices are low, producers either cut production or sell less.

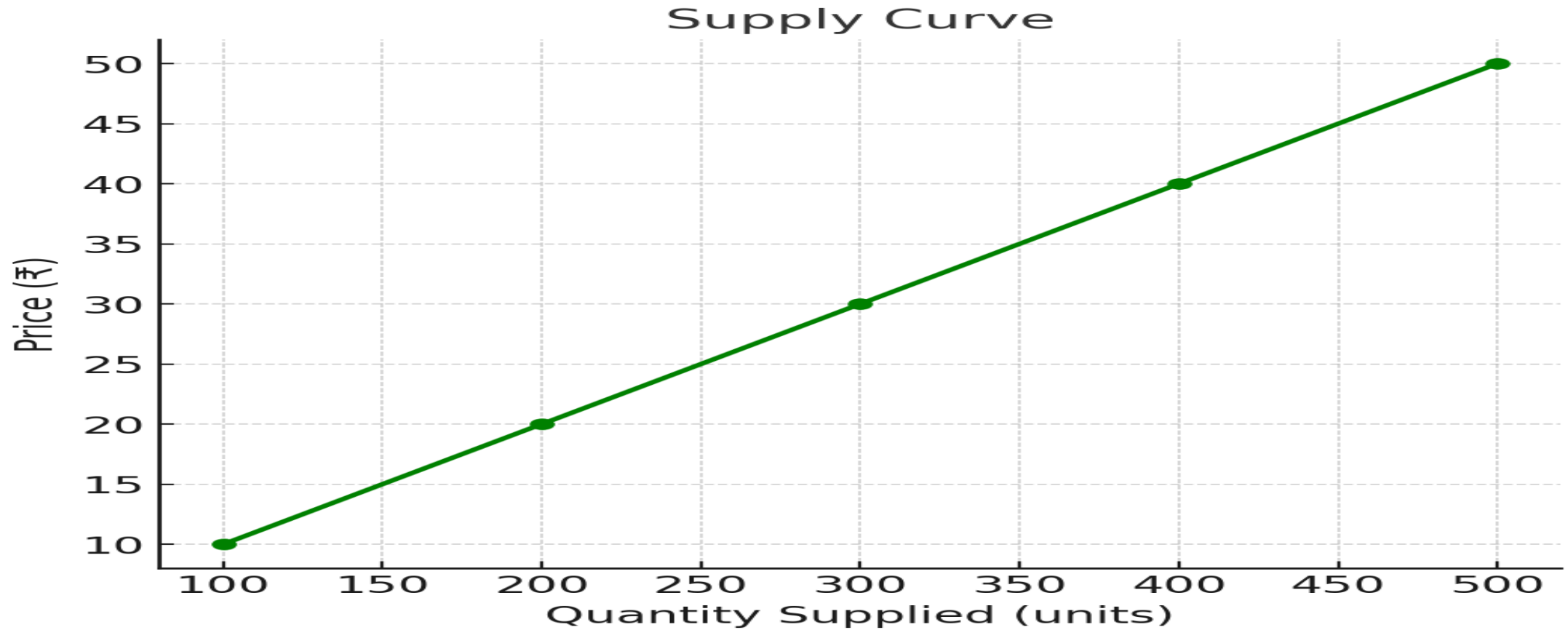
In simple words: Price $\uparrow$ $\rightarrow$ Supply $\uparrow$; Price $\downarrow$ $\rightarrow$ Supply $\downarrow$

Assumptions of the Law of Supply

- **Technology remains constant** – no sudden improvement that increases supply at the same cost.
- **Cost of production remains the same** – wages, raw material costs don't suddenly change.
- **No change in government policies** – such as taxes, subsidies, or restrictions.
- **No change in prices of related goods** – e.g., a farmer won't suddenly shift from wheat to rice due to better prices of rice.
- **No natural disturbances** – like floods, droughts, strikes, or war.
- **Perfect competition in the market** – sellers act freely without monopoly control.

Supply Curve (Diagram)

- It is an **upward sloping curve** from left to right, showing direct relation between price and supply.



- Here's the **Supply Curve** 🙌 :
- The curve slopes **upwards from left to right**, showing that when **price increases, supply also increases**.
- This confirms the **Law of Supply**.

Determinants of Supply

➤ **Price of the Commodity**

- The higher the price, the more producers are willing to supply.
- Lower prices discourage supply.
- This is the main factor explained by the **Law of Supply**.

➤ **Input Costs (Cost of Production)**

- If wages, raw material, or energy costs increase, supply decreases (because profit falls).
- If costs fall, supply increases.

➤ **Technology**

- Better technology makes production more efficient and cheaper.
- This increases supply at the same price (e.g., modern machinery in farming).

➤ **Prices of Related Goods**

- **Substitute goods in production:** If farmers get a higher price for rice, they may reduce wheat supply.
- **Joint products:** Sometimes producing one good automatically produces another (e.g., oil and oilcakes).

➤ **Expectations of Future Prices**

- If producers expect prices to rise in future, they may hold back supply now.
- If they expect prices to fall, they may sell more quickly.

➤ **Government Policy**

- **Taxes** increase production costs → reduce supply.
- **Subsidies** reduce costs → increase supply.
- **Regulations and restrictions** (like quotas or bans) also affect supply.

➤ **Natural and Climatic Factors**

- Agriculture and allied sectors heavily depend on weather, rainfall, and natural conditions.
- Natural calamities (floods, droughts, earthquakes) can reduce supply.

Market Equilibrium

➤ Meaning of Demand and Supply

Demand:

The quantity of a good or service that **consumers are willing and able to purchase** at different prices during a given period of time.

Example: At a lower price of ice cream, more students will buy ice cream.

Supply:

The quantity of a good or service that **producers are willing and able to offer for sale** at different prices during a given period of time.

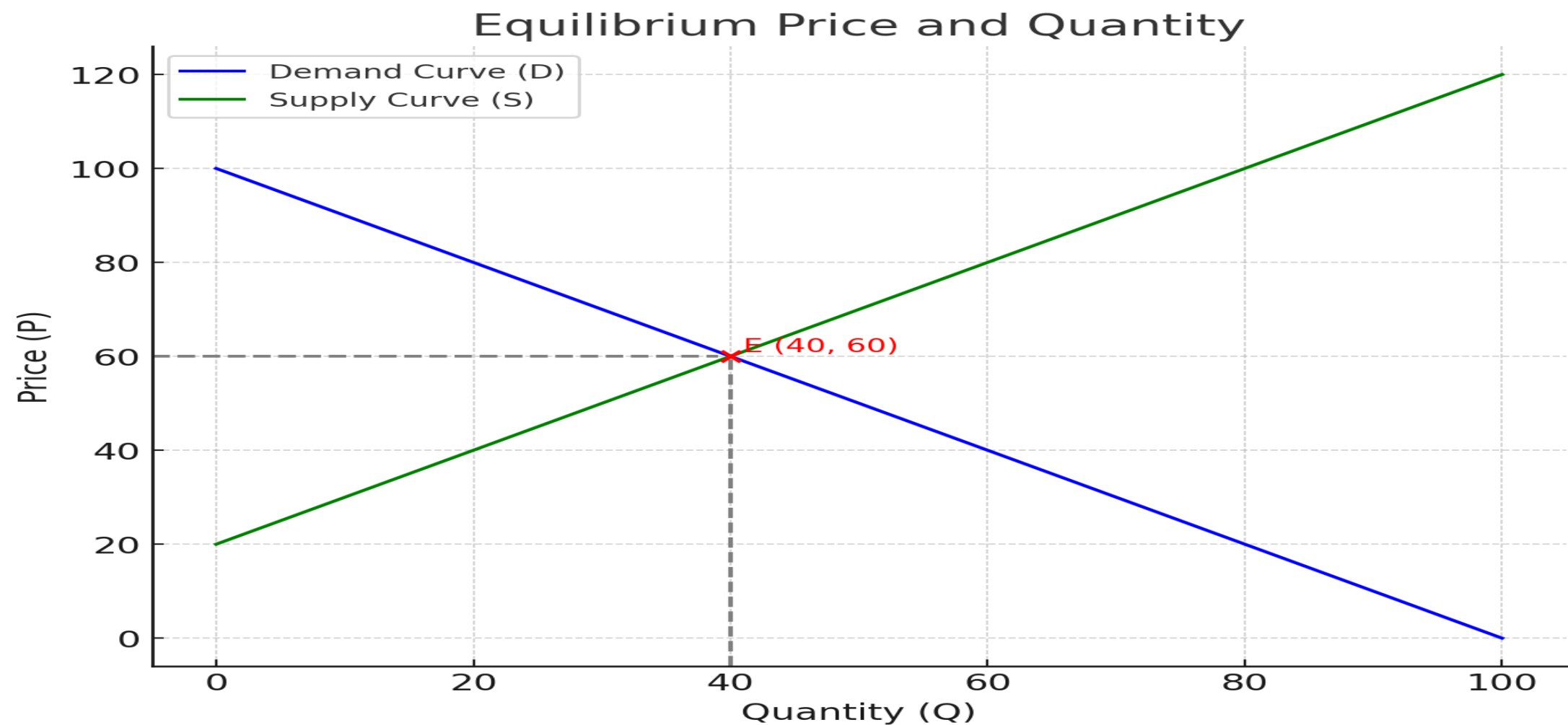
👉 Example: At a higher price of ice cream, shopkeepers are more interested in selling because they earn more profit.

- **Meaning of Equilibrium**

- In economics, **equilibrium** means a condition of **balance**.
- Market equilibrium happens when the **quantity demanded = quantity supplied** at a certain price.
- This special price is called the **equilibrium price (P^*)**.
- The quantity exchanged at this price is called the **equilibrium quantity (Q^*)**.

Determination of Equilibrium Price and Quantity

- The point where the **demand curve** and **supply curve** intersect is called the **equilibrium point**.
- The **price** at this point is the **equilibrium price (P^*)**.
- The **quantity** at this point is the **equilibrium quantity (Q^*)**.



- **Demand Curve (blue)** slopes downward.
- **Supply Curve (green)** slopes upward.
- **Equilibrium Point E (40, 60):**
 - Equilibrium Price ($P^* = 60$)
 - Equilibrium Quantity ($Q^* = 40$)

At this point, demand = supply.

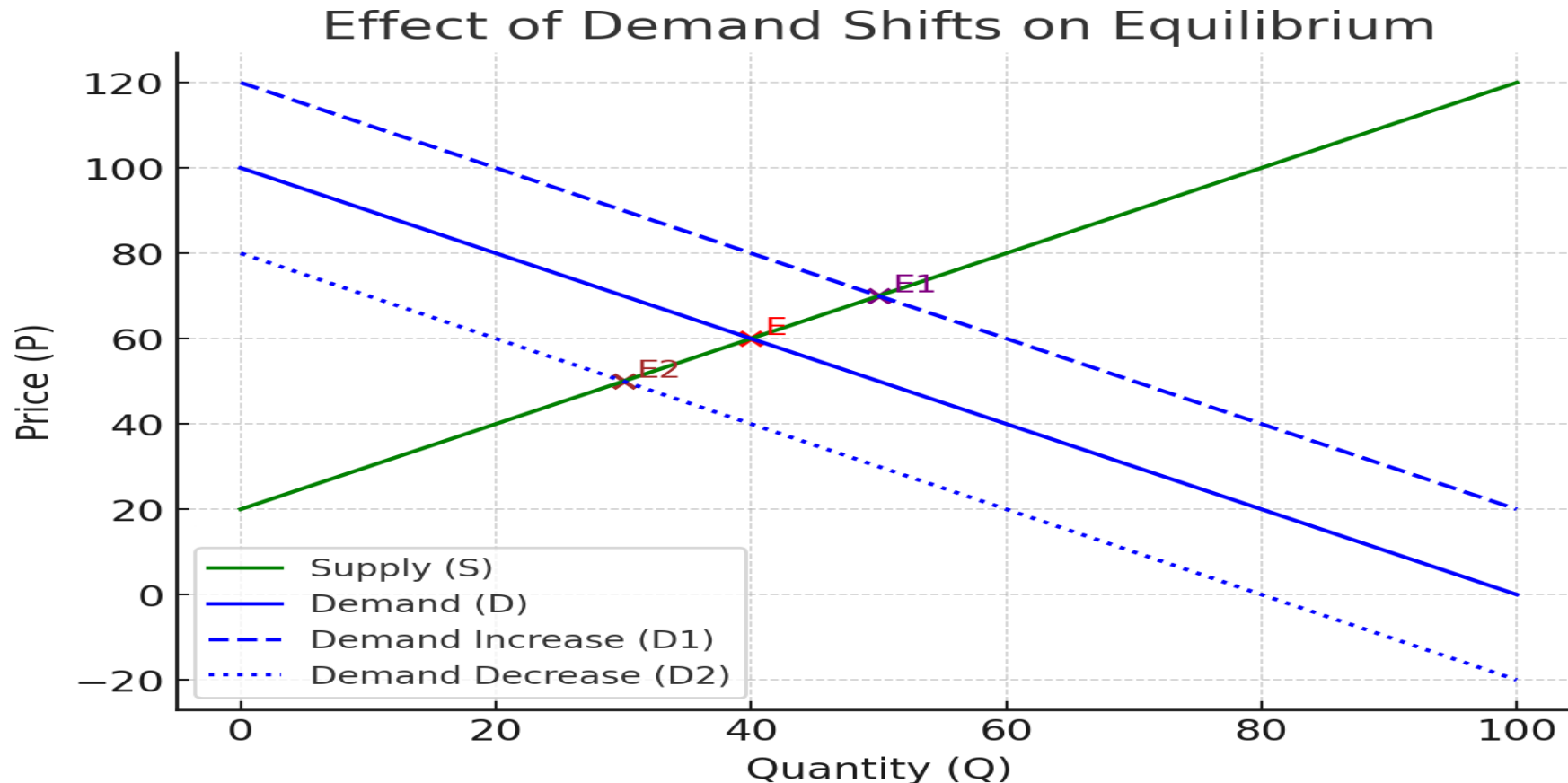
If price goes above 60 → surplus.

If price goes below 60 → shortage.

- Where these two curves cross → **Equilibrium Point (E)**.
- At **E**, price = P^* and quantity = Q^* .
- If price is **higher than P^*** → surplus (extra supply, unsold goods).
If price is **lower than P^*** → shortage (excess demand, not enough goods).

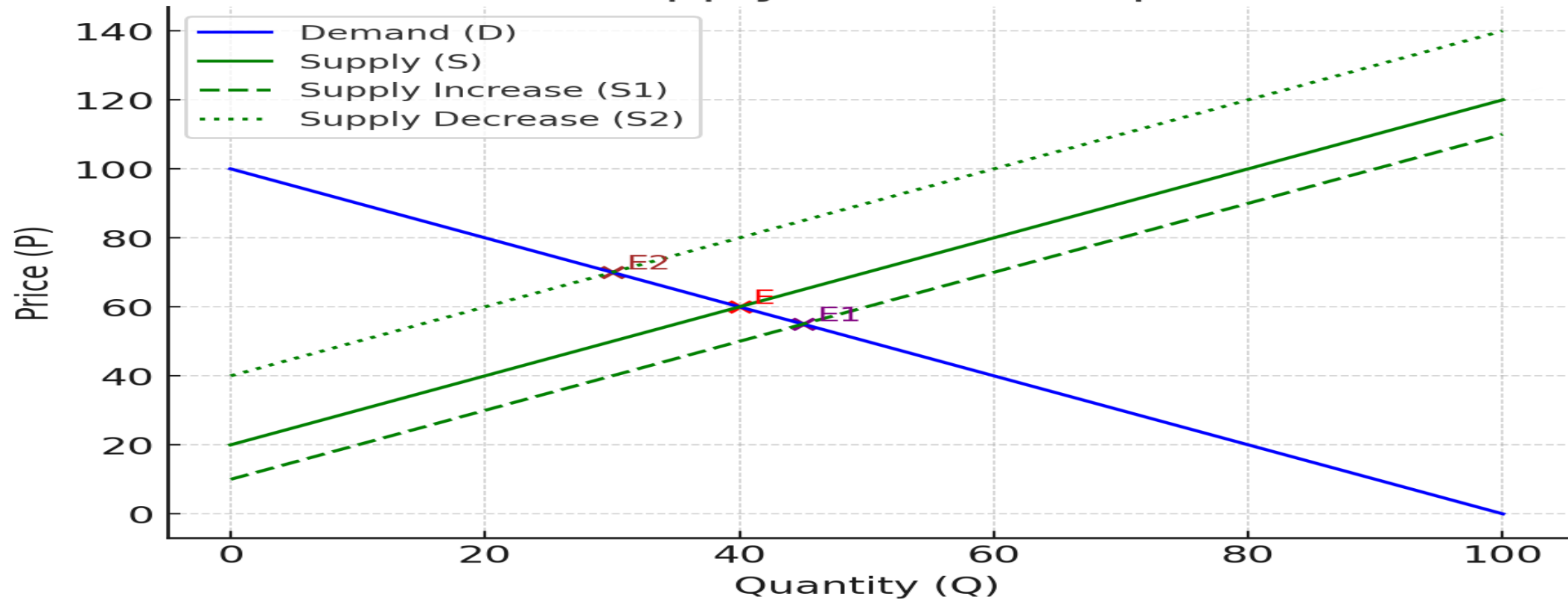
Effect of Shifts in Demand and Supply

- **Shift in Demand**
- If **demand increases** (e.g., more income, trend change, population growth), the demand curve shifts **rightward**.
New equilibrium: **higher price** and **higher quantity**.
- If **demand decreases** (e.g., fall in income, less popularity), the demand curve shifts **leftward**.
New equilibrium: **lower price** and **lower quantity**.



- Here's the diagram for **Shifts in Demand**:
- **E** = Original equilibrium.
- **E1 (purple)** = When demand **increases** → higher price & higher quantity.
- **E2 (brown)** = When demand **decreases** → lower price & lower quantity.
- **Shift in Supply**
- If **supply increases** (e.g., better technology, lower cost of production), the supply curve shifts **rightward**.
New equilibrium: **lower price** and **higher quantity**.
- If **supply decreases** (e.g., natural disaster, higher costs), the supply curve shifts **leftward**.
New equilibrium: **higher price** and **lower quantity**.

Effect of Supply Shifts on Equilibrium



- the diagram for **Shifts in Supply**:
- **E** = Original equilibrium.
- **E1 (purple)** = When supply **increases** → lower price & higher quantity.
- **E2 (brown)** = When supply **decreases** → higher price & lower quantity.

Price Elasticity of Supply

Concept of Price Elasticity of Supply

Price Elasticity of Supply (PES) measures how much the quantity of a product a company or farmer is willing to supply changes when its price changes. It tells us how flexible or responsive suppliers are to a price change.

- **Elastic supply:** If a product has an elastic supply, producers can easily make and sell a lot more of it when the price goes up. They are very flexible.
 - Example: A t-shirt manufacturer can quickly increase output by running an extra shift when the price of t-shirts rises.
- **Inelastic supply:** If a product has an inelastic supply, producers cannot easily increase the amount they sell, even if the price soars. They are not very flexible.
 - Example: A farmer cannot quickly grow more crops, like wheat or grapes, just because the price has gone up, because it takes months for them to grow.

How to measure PES...?

➤ **Percentage Method**

This is the most common and straightforward way to measure PES. You find the percentage change in the quantity supplied and divide it by the percentage change in price.

The formula is:

$$\text{PES} = \frac{\% \text{ changed in Quantity supplied}}{\% \text{ changed in Price}}$$

- Example:
- A smartphone costs ₹10,000, and a company supplies 1,000 units.
- The price increases to ₹12,000 (a 20% increase), and the company supplies 1,500 units (a 50% increase).
- $\text{PES} = 50\% / 20\% = 2.5$. Since 2.5 is greater than 1, the supply is elastic.

➤ **Arc Method**

This method is more accurate for larger price and quantity changes because it uses the average of the starting and ending values.

Example:

- Price increases from ₹100 to ₹120. Quantity supplied increases from 400 to 480.
- Percentage change in quantity supplied: $(80 / 440) \times 100 = 18.18\%$
- Percentage change in price: $(20 / 110) \times 100 = 18.18\%$
- $PES = 18.18\% / 18.18\% = 1$. This is unitary elastic, meaning the supply changes in direct proportion to the price.

➤ **Geometric Method**

- Elastic supply ($PES > 1$): The supply curve is flatter and starts from the vertical (price) axis.
- Unitary elastic supply ($PES = 1$): The supply curve goes through the origin (0,0).
- Inelastic supply ($PES < 1$): The supply curve is steeper and starts from the horizontal (quantity) axis.
- Perfectly inelastic supply ($PES = 0$): A vertical line, meaning supply never changes.
- Perfectly elastic supply ($PES = \infty$): A horizontal line, meaning suppliers will provide any amount at one specific price.

Factors Influencing Price Elasticity of Supply

- Several factors determine whether supply is elastic or inelastic:

- **Time Period**

- Immediate run → supply is inelastic (producers can't adjust output).
- Short run → moderately elastic.
- Long run → more elastic (firms can expand capacity, enter/exit industry).

- **Nature of the Commodity**

Perishable goods → inelastic supply (cannot store for long).

Durable goods → more elastic (can be stored and supplied later).

- **Availability of Factors of Production**

- If inputs are easily available, supply is elastic.
- If resources are scarce or specialized, supply is inelastic.

- **Capacity of Firms**

- Firms with idle capacity can easily increase production → elastic supply.
- **Mobility of Factors of Production**
- Greater mobility (labor, capital) → more elastic supply.
- **Storage Ability**
- Goods that can be stored (e.g., grains, steel) → more elastic supply.
- **Production Techniques**
- Simple, quick methods → elastic supply.
- Complex, time-consuming production → inelastic supply.
- **Expectations of Future Prices**
- If sellers expect higher prices later, they may withhold supply now → current supply becomes inelastic.

Income Elasticity of Demand

Income elasticity of demand (YED) measures how responsive the quantity demanded for a product is to a change in consumers' income. It tells us how consumer purchasing patterns shift as their income levels change.

The formula is:

$$\text{YED} = \frac{\% \text{ changed in quantity demand}}{\% \text{ changed in income}}$$

The result of this calculation, which can be positive, negative, or zero, helps classify goods into different categories.

1. Positive income elasticity

This applies to normal goods, where the quantity demanded increases as consumers' income rises. Most products fall into this category. Positive elasticity is further divided into three sub-types:

- Necessities (YED is between 0 and 1): The demand for these goods increases with income, but at a slower rate.
 - Example: As a person's income rises, they may buy slightly more food or electricity, but they won't increase their consumption in proportion to their income gain.

- Unitary elastic goods ($YED = 1$): The demand for these goods changes in exact proportion to the change in income.
 - Example: If income increases by 10%, the quantity demanded also increases by 10%.
- Luxury goods (YED is greater than 1): The demand for these items increases by a larger percentage than the increase in income.
- Example: When a person gets a significant raise, they might spend a much larger portion of that extra money on a new car, a vacation, or designer clothing

2. Negative income elasticity

- This applies to inferior goods, where the demand for a product decreases as a consumer's income rises. As people can afford better alternatives, they buy less of these low-cost items.
- Example: An individual who relies on cheap instant noodles or public transport may switch to healthier food options or buying a car as their income increases, causing their demand for the inferior goods to fall.

3. Zero income elasticity

- This occurs when a change in a consumer's income has no effect on the quantity demanded of a product.

Example: Essential goods like salt or basic medicines have zero income elasticity. Regardless of whether a person is rich or poor, they will buy roughly the same amount of salt.

Cross-Price Elasticity of Demand

Cross-price elasticity of demand (XED) measures how a change in the price of one product affects the quantity demanded for another product. It helps us understand the relationship between two different goods in a market.

The formula for calculating XED is:

$$\text{XED} = \frac{\% \text{ change in quantity demand of Good A}}{\% \text{ change in price of Good B}}$$

1. Substitutes (Positive XED)

- Meaning: Substitutes are products that can be used in place of each other. The cross-price elasticity for substitutes is always positive because an increase in the price of one good will cause consumers to buy more of its alternative.
- Example:
- Coffee and Tea: If the price of coffee goes up, some people will switch to buying more tea instead. Because the price of coffee and the demand for tea moved in the same direction (both went up), the XED is positive.

- Different soda brands: If the price of Coca-Cola increases, many consumers will buy more Pepsi, as it is a substitute. The demand for Pepsi rises as the price of Coca-Cola increases, leading to a positive XED.

2. Complements (Negative XED)

- Meaning: Complements are products that are consumed together. The cross-price elasticity for complements is negative because an increase in the price of one good will cause a decrease in the demand for its complementary good.

Example:

- Hot dogs and hot dog buns: If the price of hot dogs goes up, people will buy fewer hot dogs. Since they need buns for hot dogs, they will also buy fewer hot dog buns. The price of hot dogs went up, but the demand for buns went down, resulting in a negative XED.
- Printers and ink cartridges: When the price of a printer increases, fewer people will buy new printers. This naturally leads to a reduced demand for ink cartridges. The price of printers increased, but the demand for cartridges decreased, so the XED is negative

3. Unrelated goods (Zero XED)

- Meaning: Unrelated goods have no connection to one another. Changes in the price of one product have no effect on the demand for the other, resulting in a cross-price elasticity of zero.

Example:

- Pens and oranges: An increase or decrease in the price of pens will have no effect on the demand for oranges. Consumers see no connection between these two products, so the cross-price elasticity between them is zero.
- Cars and shoes: The price of a new car changing will not affect how many pairs of shoes a person buys. These are completely separate markets, so their XED is zero.

Market Failure

- A **market failure** occurs when the free market system fails to allocate resources efficiently.
- In simple words, it means the market does not produce the **right quantity of goods and services** that maximize society's welfare.
- Result: Either too much or too little of a good is produced, leading to **loss of economic efficiency**.
- Example:
- If factories pollute rivers while producing goods, the cost of pollution is not considered in the market price. Society bears the cost → market failure.

- Causes of Market Failure
- **1. Externalities**
- When the actions of producers or consumers affect third parties who are not directly involved in the market transaction.
- **Negative externalities:** Harmful effects (e.g., pollution, smoking).
- **Positive externalities:** Beneficial effects (e.g., vaccination, education).
- Market fails because these effects are not included in prices
- **2. Public Goods**
- Goods that are **non-excludable** (cannot stop others from using them) and **non-rivalrous** (one person's use does not reduce availability for others).
- Example: street lighting, national defense, public parks.
- Problem: Private firms have no incentive to produce them since they cannot charge everyone → government intervention needed.
- **3. Imperfect Competition**
- When markets are not perfectly competitive, firms may restrict output and charge high prices.
- Examples: monopoly, oligopoly, cartels.
- Leads to under-production and higher prices than in competitive markets.

- **4. Imperfect Information (Information Failure)**
- Buyers or sellers lack full knowledge about products, prices, or risks.
- Example: A patient does not know the quality of medicines, used car buyers don't know the car's true condition.
- Results in wrong choices and inefficient allocation of resources.
- **5. Factor Immobility**
- When factors of production (labour, capital, land) cannot move freely from one use or place to another.
- Example: Workers may lack skills to shift to new industries → unemployment.
- **6. Inequality in Income and Wealth**
- Free markets often create wide gaps between rich and poor.
- This can reduce overall welfare and cause social problems → considered a form of market failure.

Government Intervention

- Government intervention refers to the actions taken by the government to influence, regulate, or directly participate in the functioning of the economy. The aim is usually to correct **market failures**, ensure **fairness**, protect **consumers and workers**, and promote **economic stability**.

Causes / Need for Government Intervention

- **Market Failures** – When markets fail to allocate resources efficiently (e.g., public goods, externalities, monopolies).
- **Income Inequality** – To reduce the gap between rich and poor through taxation and subsidies.
- **Protection of Consumers and Workers** – Ensuring fair wages, product safety, and preventing exploitation.
- **Price Stability** – Controlling inflation/deflation.
- **Encouraging Economic Growth** – Through infrastructure, subsidies, and incentives.

Forms of Government Intervention

1. Price Controls

- **Price Ceiling:** Maximum price fixed by government below equilibrium price.
 - Aim: Make goods affordable (e.g., rent control, essential food items).
 - Problem: Leads to shortage, black marketing.
- **Price Floor:** Minimum price fixed by government above equilibrium price.
 - Aim: Protect producers (e.g., Minimum Support Price for farmers).
 - Problem: Leads to surplus of goods.

2. Taxes and Subsidies

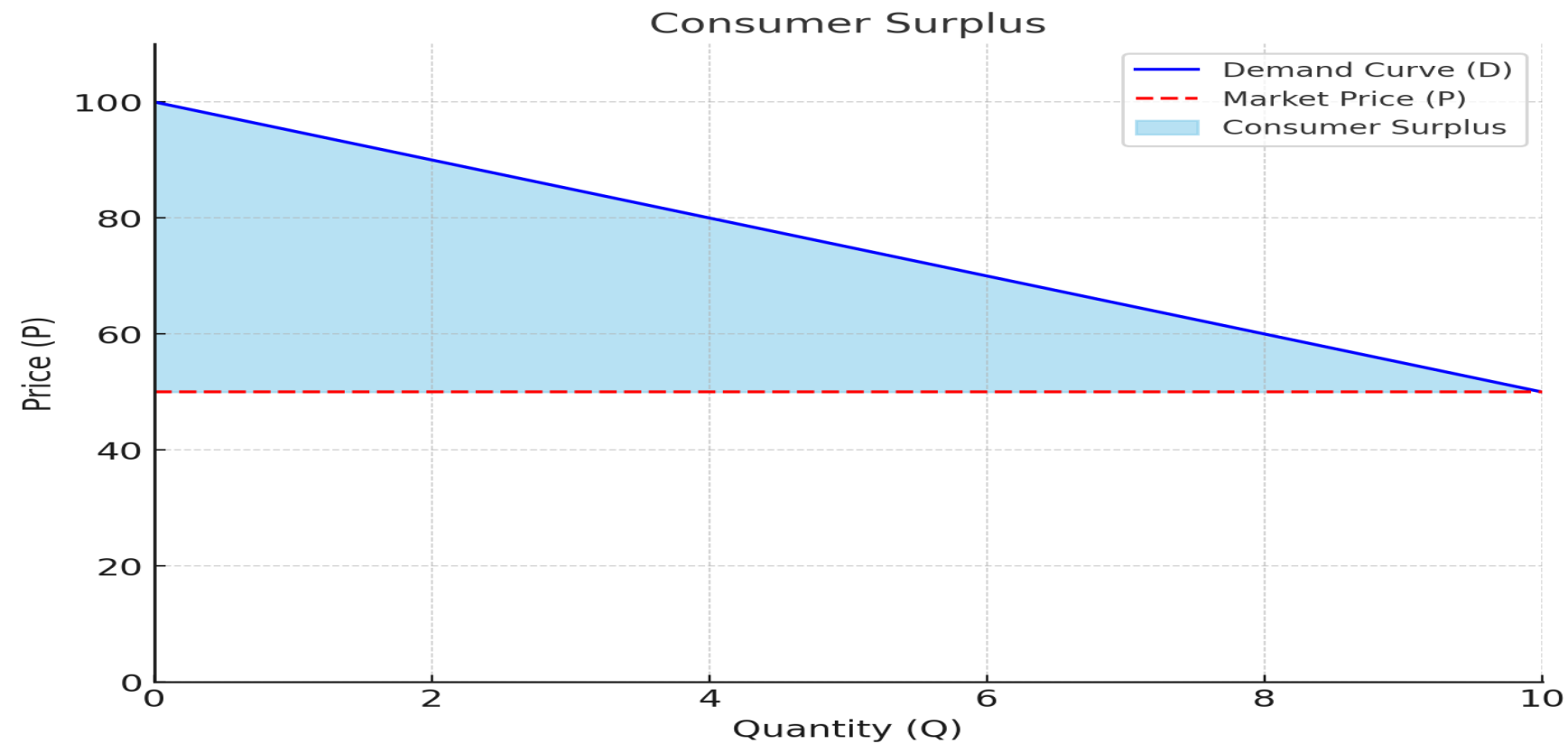
- **Taxes:** Government imposes taxes (like GST, excise duty) to reduce consumption of harmful goods (e.g., cigarettes, alcohol), raise revenue, and control inflation.
- **Subsidies:** Government gives financial support to encourage production/consumption of essential goods (e.g., fertilizer subsidy, LPG subsidy).

3. Minimum Wage Laws

- Government fixes a minimum wage that employers must pay workers.
- Aim: Protect workers from exploitation and ensure a fair standard of living.
- Effect: Raises incomes of workers, but may cause unemployment if wages are set too high.
- **Consumer Surplus**
Consumer Surplus (CS) is the **extra benefit** a consumer gets when they are willing to pay **more** for a good than the actual market price they pay.
- In simple words:
 - **Willingness to Pay (WTP)** – the maximum price a consumer is ready to pay.
 - **Market Price (P)** – the actual price in the market.
 - **Consumer Surplus = $WTP - P$**
- It shows the **gain or satisfaction** consumers enjoy because they pay less than what they were willing to.

Example

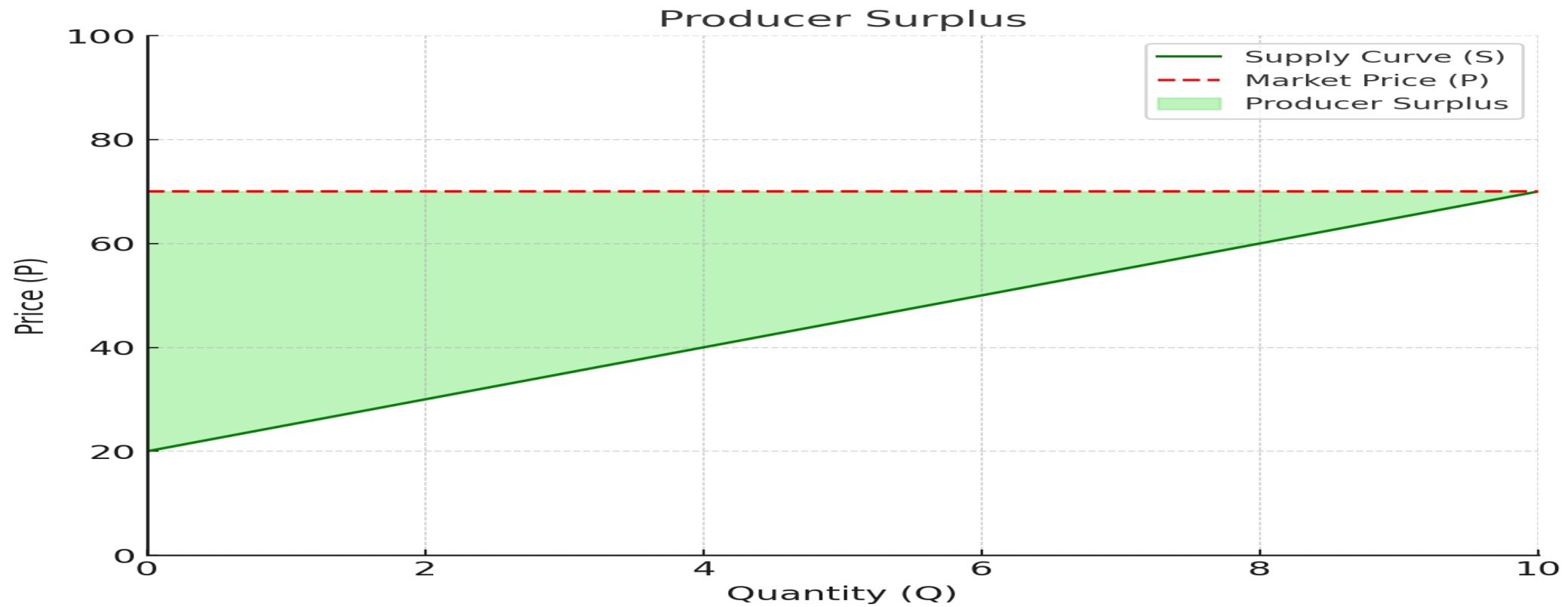
- A student is willing to pay ₹100 for a book.
- Market price is only ₹70.
- **Consumer Surplus = 100 – 70 = ₹30.**
- If 10 students buy the book with the same willingness to pay, total consumer surplus = $10 \times 30 = ₹300$.



- The **blue line** is the **Demand Curve (D)**.
- The **red dashed line** is the **Market Price (P)**.
- The **shaded sky-blue area** represents the **Consumer Surplus** → the extra benefit consumers enjoy by paying less than what they were willing to pay.

➤ Producer Surplus

- **Producer Surplus (PS)** is the extra benefit producers get when they are willing to sell a good at a **lower price**, but the market allows them to sell it at a **higher price**.
- In simple words:
 - **Market Price (P)** – actual price producers receive.
 - **Minimum Price (Supply Price)** – lowest price at which producers are willing to sell.
 - **Producer Surplus = P – Supply Price**
- It shows the **gain or profit** producers earn by selling at a price higher than their minimum acceptable price.
- **Example**
- A farmer is willing to sell wheat at **₹800 per quintal**.
- Market price is **₹1,200 per quintal**.
- **Producer Surplus = 1,200 – 800 = ₹400.**
- If 10 quintals are sold, total producer surplus = $10 \times 400 = \text{₹4,000}$.

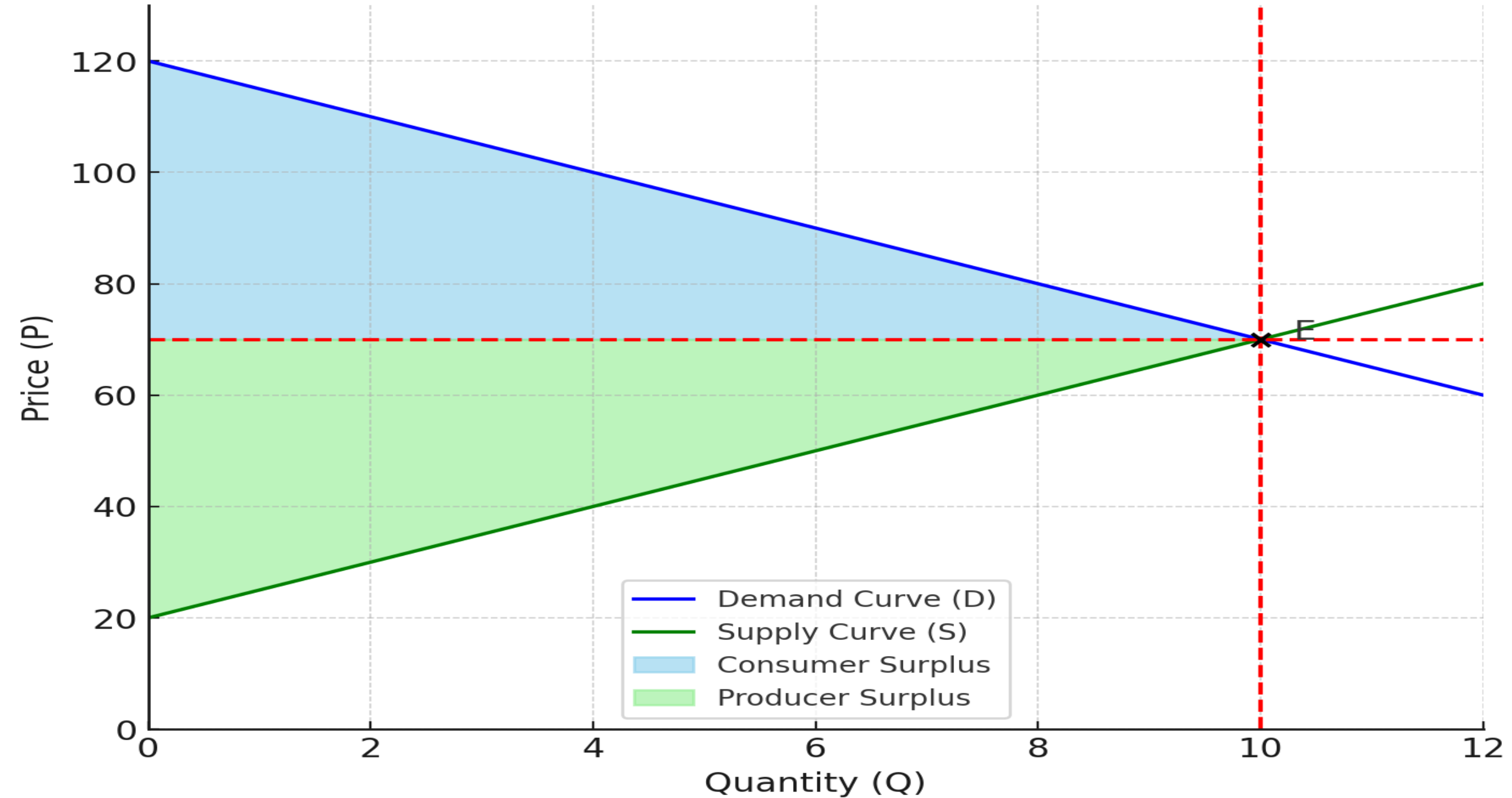


- Here's the **Producer Surplus** diagram 🙋
- The **green line** is the **Supply Curve (S)**.
- The **red dashed line** is the **Market Price (P)**.
- The **shaded light-green area** shows the **Producer Surplus** → the extra benefit producers get when they sell at a higher price than their minimum acceptable price.

Efficiency in Allocation of Resources

- **Efficiency in allocation of resources** means that the limited resources of an economy are used in such a way that **maximum satisfaction for consumers** and **maximum profit for producers** is achieved.
- In other words, **resources are allocated efficiently when goods and services are produced in the right quantities and distributed to those who value them the most.**
- This happens at the **market equilibrium** (where demand = supply).
- **Conditions for Efficient Allocation**
- **Consumer Side (Consumer Surplus)**
 - Consumers buy goods at a price **less than what they are willing to pay.**
 - This creates **consumer surplus** (extra benefit for consumers).
- **Producer Side (Producer Surplus)**
 - Producers sell goods at a price **higher than their minimum acceptable cost.**
 - This creates **producer surplus** (extra benefit for producers).
- **Total Surplus (Economic Welfare)**
 - **Total Surplus = Consumer Surplus + Producer Surplus**
 - At equilibrium, **total surplus is maximized** → no resources are wasted, and no one can be made better off without making someone else worse off.

Efficiency in Allocation of Resources



Here's the **Efficiency in Allocation of Resources** diagram 🙌

- **Blue curve** → Demand (D)
 - **Green curve** → Supply (S)
 - **Black point (E)** → Market Equilibrium (efficient allocation point)
 - **Sky-blue area** → **Consumer Surplus** (extra benefit to consumers)
 - **Light-green area** → **Producer Surplus** (extra benefit to producers)
 - At equilibrium, **total surplus (consumer + producer surplus) is maximized**, meaning resources are allocated **efficiently** with no waste.
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- Efficiency in allocation of resources means that at equilibrium:
 - Consumers get maximum satisfaction,
 - Producers get maximum profit,
 - Society achieves **maximum total welfare**.